

Suggested Teaching Guideline for
Network Defense & Countermeasures PG-DITISS August 2025

Duration: 30 class room hrs + 40 lab hrs (Total: 70 Hrs)

Objective: To introduce the students to Network Defense and Countermeasures.
This includes the following:

- Network Security Concepts,
- Firewalls,
- IDS & IPS, and
- VPN

Prerequisites: OS and Network Concepts

Evaluation method:

Theory Exam: 40% weightage.

Lab Exam: 40%weightage.

Internal Assessment: 20%weightage.

List of books / Other training Material:

Courseware: Cryptography & Network Security: Principles and Practices by William Stallings.

Reference:

- Fundamentals of network and Security: Eric Maiwald/TMH

Note: Every Session includes 2 Hours of theory and 2 Hours of Lab, unless indicated Otherwise.

Session 1:

- Introduction to Information Security
- Why Information Security?
- Security: The money factor involved
- Internet Statistics - Study from a security perspective
- Vulnerability, Threat and Risk
- Qos

Lab 1:

- Install and configure an antivirus software to scan your system.
- Analyze and summarize internet security statistics from OWASP or other reliable sources.
- Conduct a vulnerability scan on your local network using OpenVAS.

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- Simulate network traffic and measure QoS parameters using Wireshark.

Session 2:

- Risk Management, Exposure and Countermeasure
- Firewall
- De-militarized Zone
- Two methods of implementing firewall

Lab 2:

- Identify and categorize risks in your network using the NIST Cybersecurity Framework.
- Install and configure pfSense as a firewall on your network.
- Set up a DMZ using a virtual environment to host a web server with pfSense.
- Configure and compare packet-filtering and proxy firewalls using iptables and Squid.

Session 3:

- Packet Filtering
- Screened Host Firewall
- Stateful Inspection Firewall
- NextGen Firewall app controls
- iptables - Linux Firewall

Lab 3:

- iptables - Rule Processing
- Default Policy
- iptables - Predefined tables
- INPUT
- OUTPUT
- FORWARD

Lab 4:

- Iptables / netfilter/ Xtables-Addons
- verifying iptables / netfilter
- internal working
- iptables / netfilter - concept of targets

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- DROP
- ACCEPT
- iptables - SPI Firewall
- Setting up a SPI Firewall - Standard installation
- Iptables IPv6 Rule Management

Lab 5:

- Automating iptables and scripting
- Bash Scripting to automate iptables
- Advanced iptables
- Loading of modules
- Geo IP Blocking using Xtables-Addons

Lab 6:

- Access control using iptables
- Internet sharing Using Iptables

Session 4:

- Wireshark
- Create a filters for data collection and display
- Examine real-world packet captures

Lab 7:

- Wireshark
- Examine real-world packet captures
- Iptables Port Forwarding
- Iptables use case: Fail2ban

Session 5:

- Linux Software Firewall(ClearOS Pfsense)
- Nginx & Squid Reverse Proxy
- UTM
- Server Load Balancing

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Lab 8:

- Nginx & Squid Reverse Proxy
- Configure reverse proxy URL using regex
- Server Farming

Session 6 & 7:

- VPN – Introduction
- VPN protocols/characteristics
- VPN Functions
- Types of VPN
- SecureVPN
- Trusted VPN

Lab 9:

- OpenVPN configuration in both Linux & Windows
- Site to Site Connectivity
- Certificate & Password dependent authentication
- VPN configuration for Mobile Device

Lab 10:

- Pfsense OS - Installation and Configuration
- Installation of Pfsense OS and Basic
- Configuration

Lab 11:

- UTM Configuration
- Basic Setup of UTM

Lab 12:

- Configuration for access control and Firewall features (UTM)
- UTM VPN Configuration

Session 8:

- Hybrid VPN
- IPsec
- Tunnel mode/transport mode

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- Ipv6 VPN
- Split Tunnel full tunnel VPN

Lab 13:

- VPN Configuration under Windows 2016 using RRAS
- L2TP/PPTP VPN Setup

Session 9:

- Introduction to IDS and IPS
- IDS / IPS
- Types of Attacks
- IDS
- Security Events
- Vulnerability/design/implementation

Lab 14:

- Distributed Honeynet System (Developed by C-DAC)
- Dynamically configure Honeypot
- UAC (URL Analyzer and Classifier)
- Tcpdump installation, verification and basic usage of tcpdump

Session 10 & 11:

- Attacks-traditional/distributed
- Intruder types
- Types of IDS
- IPS categories
- Defence in depth
- IDS and IPS analysis scheme
- Detection methodologies
- Principles of IDS
- Threat Hunting model

Lab 15:

- Install and compare Snort (NIDS) and OSSEC (HIDS) on your network.
- Set up Suricata as an IPS and configure rules for network intrusion prevention.

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Lab 16:

- Analyze network traffic logs from Snort and Suricata to identify and respond to threats.
- Perform threat hunting using the ELK stack (Elasticsearch, Logstash, Kibana) to detect anomalies.

Session 12 & 13:

- Symptoms of attacks
- Tired architecture
- Sensors-network/host based
- Denial of services
- Dos & DDos Mitigation
- Sensor Deployment
- Agents
- Functions of IDS agents
- IDS Manager

Lab 17:

- Identify and document symptoms of various attacks using Wireshark.
- Design and implement a three-tier architecture in a virtual environment.
- Set up and configure network-based sensors with Snort and host-based sensors with OSSEC.

Lab 18:

- Simulate a DoS attack using LOIC and analyze its impact on a target system.
- Implement rate limiting and IP blacklisting using iptables to mitigate DoS/DDoS attacks.

Session 14:

- Introduction of Log Analyser
- Log
- SIEM Log Correlation and event triggering
- Introduction of SIEM
- SIEM Log Forwarding Configuration
- SIEM Log Correlation and event triggering

Lab 19:

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- Snort
- Writing Basic Snort Rules
- Syslog Server
- BASE
- Configuration and deployment of SIEM

Session 15:

- Testing Snort in both Windows & Linux
- IDS architecture
- Bypassing an IDS

Lab 20:

- Testing of Snort using a simulated attack
- Nagios
- Nagios Sensor Configuration (Windows & Linux)
- Nagios with Email services
- Nagios with up/Down of Ram/Hard Disk